

Uni. Roll No.

Program/ Course: **B.Tech.** (Batch 2018 onward)Semester: **3rd**Name of Subject: **Digital Electronics**Subject Code: **ESCS-101**Paper ID: **16012****Time Allowed: 03 Hours****Max. Marks: 60****NOTE:**

- 1) Parts A and B are compulsory
- 2) Part-C has Two Questions Q8 and Q9. Both are compulsory, but with internal choice
- 3) Any missing data may be assumed appropriately

Section – A**[Marks: 02 each]****Q1.**

- a) Differentiate between latch and flip-flop.
- b) Explain Field Programmable Gate Arrays.
- c) Differentiate between Mealy and Moore circuits.
- d) What is 'Race around Condition'? How it can be eliminated?
- e) Convert the following: (i) $(78.12)_{10} = (?)_2$, (ii) $(1011111011)_2 = (?)_{\text{gray-code}}$
- f) Simplify the following Boolean expression $(AB)' + A' C + A' B' C'$.

Section – B**[Marks: 04 each]****Q2.** Explain PLA and PAL circuits with examples.**Q3.** Explain the working of master-slave JK flip-flop with neat diagram.**Q4.** Design a 4-bit PISO shift register.**Q5.** Using K-map, minimize the given Boolean function

$$F(A, B, C) = \bar{A}C + \bar{A}B + A\bar{B}C + BC.$$

Implement the simplified circuit using NAND gates only

Q6. Design a 3-bit Gray to Binary code convertor and draw its' circuit.**Q7.** State and prove De-Morgan's theorem.

EVENING

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Section – C [Marks: 12 each (06 for each sub-part, if any)]

Q8. Explain the working of standard TTL NAND gate. Compare TTL and CMOS logic family characteristics.

OR

Give various types of A/D converters. Explain their performance characteristics.

Q9. (a) Implement $Y = \pi M(1,5,8,11,12)$ using MUX

(b) Explain various binary codes with examples.

OR

Design a circuit that will generate a odd parity bit for 4 bit input and implement it using only XNOR gates.
